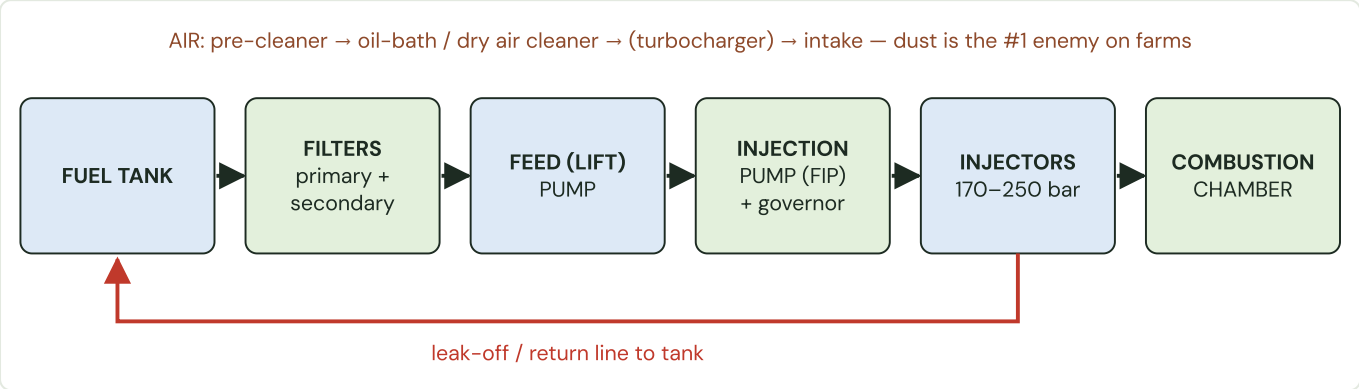
UNIT 4 · TOPIC 2(B)

Fuel, Air and Governing Systems

Syllabus topic: Various systems of IC engines: operation, adjustment and troubleshooting



Diesel fuel and air supply system (redrawn from the source figure)

Fuel system components, in flow order

1. Gravity or lift-pump feed.
2. Paper/cloth **primary and secondary filters** (with water separator).
3. **Fuel injection pump (FIP):** inline (one plunger-barrel per cylinder; the plunger HELIX meters quantity via the control rack) or distributor (rotary — one element serves all cylinders). Delivery valves maintain line pressure. The mechanical flyweight GOVERNOR is mounted on the FIP.
4. **Injectors:** multi-hole (direct injection) or pintle (indirect injection); opening pressure ~**170–250 bar** (conventional).

Test injectors on a nozzle tester for: opening pressure, spray pattern, chatter and nozzle drip.

Air system

- Air cleaning is critical in dusty field work: a pre-cleaner (cyclone bowl) + an OIL-BATH air cleaner (older) or a DRY paper element with dust unloader (modern).
- A choked air cleaner → rich mixture → black smoke and power loss.
- A turbocharger (exhaust-gas driven) + intercooler raises power ~**20–30%** from the same displacement.

Adjustments and service

Bleeding (priming) — after a filter change or a fuel run-out:

1. Loosen the bleed screws.
2. Operate the primer until bubble-free fuel flows.
3. Tighten the screws. Skipping this leaves an air lock that prevents starting.

- **FIP calibration and phasing** on a test bench: delivery per 100 strokes, within inter-cylinder variation limits.
- **Injection timing:** set by the spill method.
- **Injector pressure:** set by shims or adjusting screw.

Electronic common-rail (CRDI) injection is replacing mechanical pumps on larger tractors — see Topic 7b.

Governing system

Purpose: hold the set speed as the load varies.

Method	How	Used on
QUALITY governing	Vary fuel per stroke; air unthrottled	CI (diesel) engines
QUANTITY governing	Throttle the mixture	SI engines
Hit-and-miss	Skip firing	Old single-cylinder engines

- Tractor engines use **variable-speed centrifugal flyweight governors** — any set speed via the hand throttle.
- **Governor DROOP (regulation):** the small speed fall from no-load to full-load, as %.
- **HUNTING:** rhythmic surging caused by over-sensitive or sticking linkage.

Practice MCQs

Q46. In an inline FIP, fuel quantity is metered by:
(a) Throttle plate
(b) Rotating plunger helix via the control rack
(c) Venturi
(d) Float

Q47. Conventional diesel injector opening pressure is about:
(a) 20–50 bar
(b) 170–250 bar
(c) 1000–1500 bar
(d) 5 bar

Q48. After changing fuel filters, a diesel must be:
(a) Timed
(b) Bled of air (primed)
(c) Rebored
(d) Decoked

Q49. An injector is bench-tested for:
(a) Opening pressure, spray pattern, chatter, drip
(b) Colour
(c) Weight
(d) Torque

Q50. A choked air cleaner produces:
(a) White smoke
(b) Black smoke and power loss
(c) Blue smoke
(d) No effect

Q51. The oil-bath air cleaner is being replaced by:
(a) No cleaner
(b) Dry paper element with cyclonic pre-cleaner
(c) Water bath
(d) Sponge only

Q52. A turbocharger is driven by:
(a) Crankshaft belt
(b) Exhaust gases
(c) Camshaft
(d) PTO

Q53. Diesel engines use which method of governing?
(a) Quantity (throttle)
(b) Quality (vary fuel, air unthrottled)
(c) Hit and miss
(d) Spark control

Q54. Rhythmic surging of engine speed due to over-sensitive governor is:
(a) Droop
(b) Hunting
(c) Knocking
(d) Dieseling

Q55. Governor droop (regulation) is the:

(a) Oil leak
(b) Speed fall from no-load to full-load as %
(c) Belt slip
(d) Rack travel

Answers

Q46 — (b) Rotating plunger helix via the control rack
Q47 — (b) 170–250 bar
Q48 — (b) Bled of air (primed)
Q49 — (a) Opening pressure, spray pattern, chatter, drip
Q50 — (b) Black smoke and power loss
Q51 — (b) Dry paper element with cyclonic pre-cleaner
Q52 — (b) Exhaust gases
Q53 — (b) Quality (vary fuel, air unthrottled)
Q54 — (b) Hunting
Q55 — (b) Speed fall from no-load to full-load as %

Sources: Liljedahl et al. — Tractors and their Power Units; Jagdishwar Sahay; manufacturer service practice; TREM notifications; NPTEL farm power lectures.